ALICE L WALTON SCHOOL OF MEDICINE

Antiarrhythmic Agents: Classes II–IV

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Adapted from: Antiarrhythmic Drugs (Classes II, IV, and V)

Listen to this Brick

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Learning Objectives (3) Versions

After completing this brick, you will be able to:

1. Explain the mechanism of action of antiarrhythmic agents (Class II, III, and IV) and digoxin.
2. Explain the changes in action potential and the associated ECG changes with digoxin and Class II, III, IV antiarrhythmic agents.
3. Recognize th ents (Class II, III, and IV) and digoxin.

CASE CONNECTION

After office hours, you return a call to JG, a 59-year-old male with newly diagnosed atrial fibrillation. "If someone is going to get a side effect to a medication, it's going to be me," JG begins. "Ever since my cardiologist gave me that new pill, my bowels have really been giving me a problem. I don't have a bowel movement for about 5-6 days. I'm tempted to just stop it. Do you think that's OK?" "Let's think about this together," you reply. "By any chance, is that new medication verapamil?" "Sure is. How'd you know that? Is there something else I can take?"

What is the mechanism of action of verapamil in the treatment of JG's atrial fibrillation? What screening questions would you ask him before prescribing this medication? Consider your answers as you read, and we'll revisit JG at the end of the brick.

GO TO CONCLUSION ↓

What Are Antiarrhythmic Drugs?

Antiarrhythmic agents are classified based on the Vaughan-Williams classification. We have discussed Class I agents in the Anti-arrhythmic agents: Class I brick.

Mnemonic

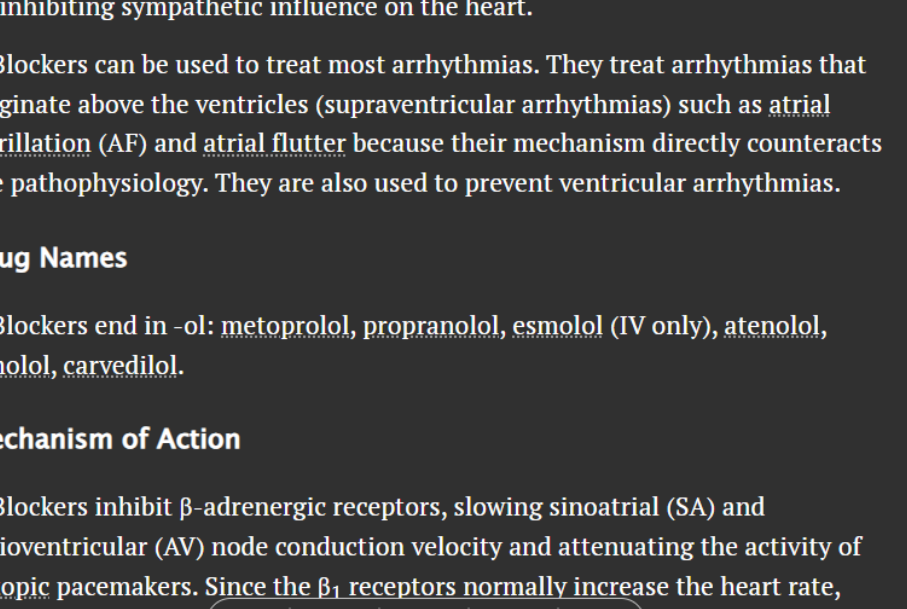
Order of the Vaughan-Williams classification = **S**ome **B**lock **P**otassium **C**hannels: **S**odium, **β-B**lockers, **P**otassium, and **C**alcium.

Here, we'll focus on classes II, III and IV antiarrhythmic drugs, which block the action potential of cardiac pacemaker cells. We'll also cover a few antiarrhythmic drugs that do not fit into the Vaughan-Williams classification.

Normal Pacemaker Cell/Slow-Type Action Potential

Classes II and IV drugs act mainly by affecting the **cardiac pacemaker cell action potentials**. Before describing their blocking effects in more detail, let's review the normal action potential of pacemaker cells (Figure 1):

- Phase 4: Gradual depolarization of the pacemaker cell is elicited by movement of the sodium current (called the funny current [I_f]) through hyperpolarization-activated cyclic nucleotide-gated (HCN) channels until the threshold potential is reached.
- Phase 0: Depolari nflux through the L-type calcium cha
- Phase 3: Repolarization is caused by the delayed outward potassium (K^+) current (I_K).



QUIZ

Figure 1 Action potential of pacemaker cells

Note that the heart rate is determined by the slope of the phase 4 of this action potential—a slower depolarization leads to slower heart rate, and a faster depolarization leads to a faster heart rate. The action potential of the pacemaker cells varies from the cardiac myocytes in several respects.

CLINICAL CORRELATION

Drugs such as acetylcholine and adenosine decrease the rate of depolarization and heart rate, whereas catecholamines (e.g., epinephrine) have the opposite effect.

What Are β-Blockers (Class II Antiarrhythmics)? (LO#1–3)

Class II antiarrhythmic agents, or β-blockers, exert their antiarrhythmic effect by inhibiting sympathetic influence on the heart.

β-Blockers can be used to treat most arrhythmias. They treat arrhythmias that originate above the ventricles (supraventricular arrhythmias) such as atrial fibrillation (AF) and atrial flutter because their mechanism directly counteracts the pathophysiology. They are also used to prevent ventricular arrhythmias.

Drug Names

β-Blockers end in -ol: metoprolol, propranolol, esmolol (IV only), atenolol, timolol, carvedilol.

Mechanism of Action

β-Blockers inhibit β-adrenergic receptors, slowing sinoatrial (SA) and atrioventricular (AV) node conduction velocity and attenuating the activity of ectopic pacemakers. Since the $β_1$ receptors normally increase the heart rate, these agents **slow th** g the sympathetic nervous system, which would otherwise increase SA and AV node activity and conduction velocity.

On a cellular level, β-blockers slow (decrease the slope of) phase 4 of the nodal (pacemaker) action potential, delaying depolarization (Figure 2, blue curve). They will reduce the conduction velocity through the AV node and shorten action potential duration (APD), primarily by decreasing Ca^{2+} influx.

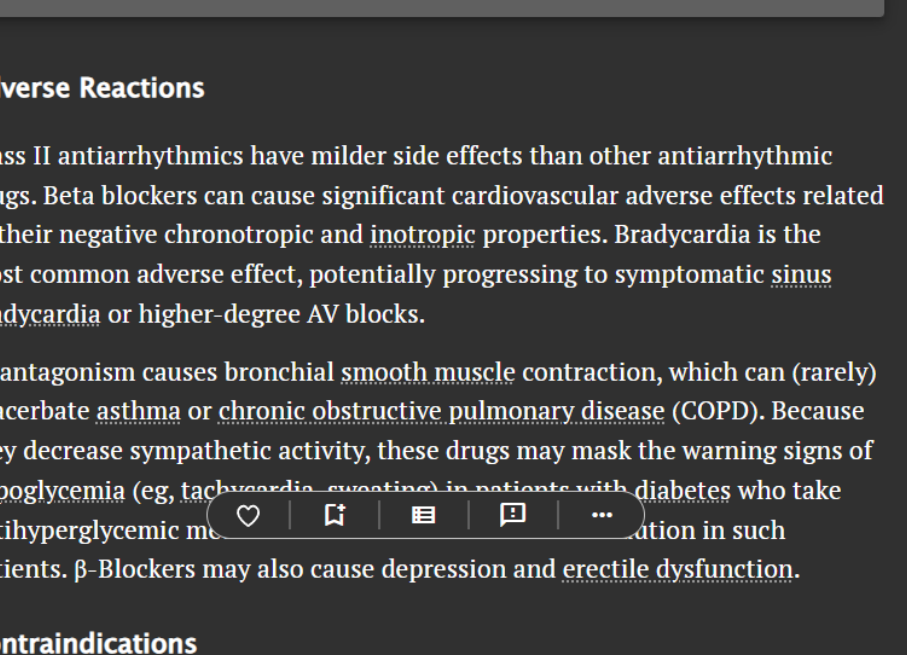


Figure 2 Effect of class II antiarrhythmics on pacemaker action potential

Effect on Electrocardiogram

The delayed depolari **the PR interval** on the ECG, effectively causing a drug-induced first-degree AV block.

Why are β-blockers used to treat supraventricular arrhythmias?

Adverse Reactions

Class II antiarrhythmics have milder side effects than other antiarrhythmic drugs. Beta blockers can cause significant cardiovascular adverse effects related to their negative chronotropic and inotropic properties. Bradycardia is the most common adverse effect, potentially progressing to symptomatic sinus bradycardia or higher-degree AV blocks.

β2 antagonism causes bronchial smooth muscle contraction, which can (rarely) exacerbate asthma or chronic obstructive pulmonary disease (COPD). Because they decrease sympathetic activity, these drugs may mask the warning signs of hypoglycemia (eg, tach diabetes who take antihyperglycemic me ation in such patients. β-Blockers may also cause depression and erectile dysfunction.

Contraindications

β-Blockers are contraindicated in 2nd and 3rd degree heart blocks.

What Are Calcium-Channel Blockers (Class IV Antiarrhythmics)? (LOs#1–3)

Class IV antiarrhythmics are the Ca^{2+} channel blockers (CCBs). Class IV antiarrhythmics, primarily verapamil and diltiazem, selectively block L-type calcium channels in cardiac tissue. These act on nodal (pacemaker) cells as well as on cardiac myocytes.

Indications

Class IV drugs are used to reduce the heart rate in some supraventricular tachycardias such as AF and atrioventricular node reentrant tachycardia (AVNRT).

Drug Names

Class IV drugs include **verapamil and diltiazem** (note that the dihydropyridine CCBs such as nifedipine are not used as antiarrhythmic drugs).

Mechanism of Action

Ca^{2+}channel blockers block the rapid influx of Ca^{2+} ions in phase 0 that is responsible for depol ion potential by prolonging depolarization and repolarization, mainly at the AV node, a process called negative dromotropy (Figure 3, blue curve).

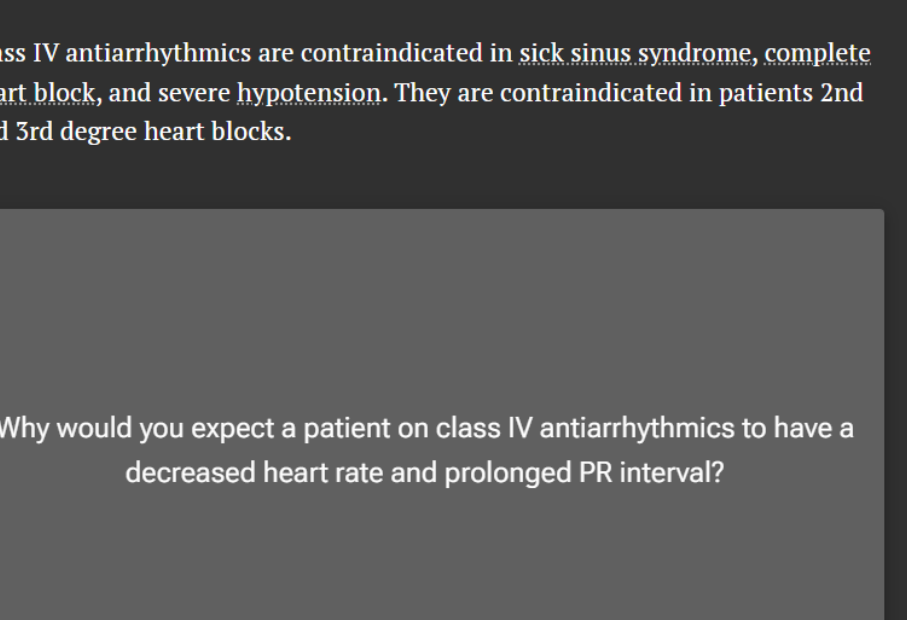


Figure 3 Effect of class IV antiarrhythmics on pacemaker action potential

This prolongation increases both the effective refractory period and the length of transmission time from atria to ventricles; hence slowing down the conduction velocity. Thus, these drugs decrease the firing rate of aberrant pacemaker sites within the heart.

Effect on Electrocardiogram

Verapamil and diltiaz onged PR interval.

Adverse Reactions

Class IV antiarrhythmics may cause constipation and peripheral edema. This makes sense because Ca^{2+} is required for bowel peristalsis, venous contraction, and the pacemaker action potential. Cardiac adverse effects include excessive bradycardia, AV block progression, and negative inotropic effects that can precipitate or worsen heart failure.

Contraindications

Class IV antiarrhythmics are contraindicated in sick sinus syndrome, complete heart block, and severe hypotension. They are contraindicated in patients 2nd and 3rd degree heart blocks.

Why would you expect a patient on class IV antiarrhythmics to have a decreased heart rate and prolonged PR interval?

What Are Potassium-Channel Blockers (Class III Antiarrhythmics)? (LO#1–3)

Class III antiarrhythmics block K^+ channels in cardiac myocytes, affecting primarily phase 3 of the cardiac myocyte action potential. These drugs are used to treat atrial arrhythmias such as AF and atrial flutter and ventricular arrhythmias such as VT. Class III drugs show reverse use dependence, meaning they are more active at slow heart rates because they have higher affinity for resting channels.

Drug Names

Some class III antiarrhythmics include amiodarone, ibutilide, dofetilide, and sotalol.

Mechanism of Action

Class III drugs block K^+ channels during phase 3, slowing the efflux of K^+. This **delays the repolarization** of myocytes, **increasing the duration of both the action potential and effective refractory period** (Figure 4).

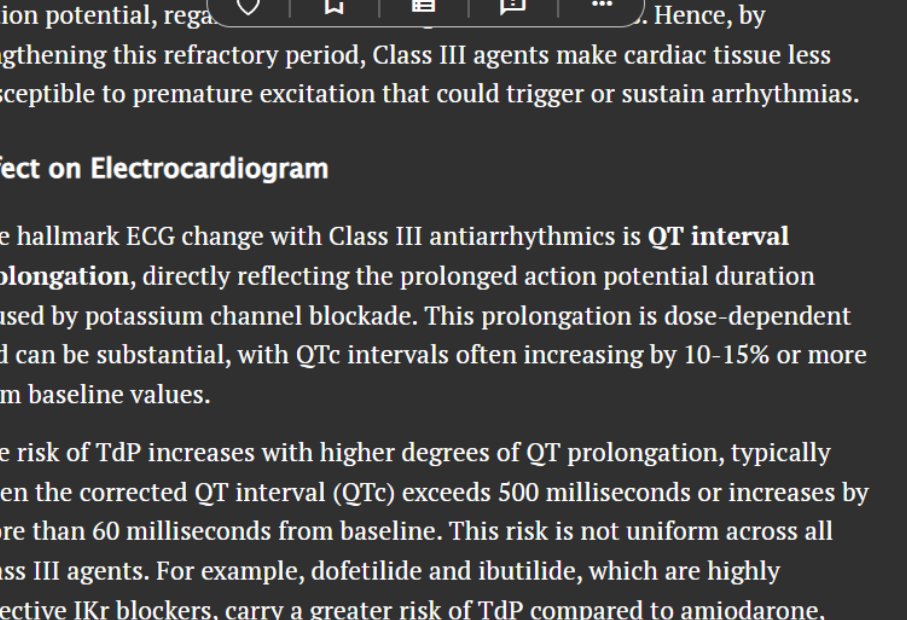


Figure 4; Class III Antiarrhythmic agents action potential

Specifically, these agents inhibit the rapid component of the delayed rectifier potassium current (IKr), which is mediated by the human ether-a-go-go-related gene (hERG) potassium channel. Some agents also affect the slow component (IKs) and other potassium currents to varying degrees. By blocking these outward potassium currents, Class III agents prevent the **efflux** of positively charged potassium ions from cardiac cells during repolarization.

The potency and selectivity of potassium channel blockade varies among different Class III agents. For example, dofetilide and ibutilide are highly selective IKr blockers, while amiodarone has a broader spectrum of activity, affecting multiple ion channels including sodium and calcium channels, giving it properties of other antiarrhythmic classes as well. Sotalol exhibits both Class III effects (potassium channel blockade) and Class II effects (beta-adrenergic receptor blockade).

As shown in Figure 4, by blocking potassium channels, Class III antiarrhythmics prolong phase 3 (repolarization) of the cardiac action potential. This prolongation increases the APD and consequently extends the effective refractory period (ERP) of cardiac tissue. Recall that the ERP is the period during which cardiac cells cannot be re-excited to generate another action potential, rega . Hence, by lengthening this refractory period, Class III agents make cardiac tissue less susceptible to premature excitation that could trigger or sustain arrhythmias.

Effect on Electrocardiogram

The hallmark ECG change with Class III antiarrhythmics is **QT interval prolongation**, directly reflecting the prolonged action potential duration caused by potassium channel blockade. This prolongation is dose-dependent and can be substantial, with QTc intervals often increasing by 10-15% or more from baseline values.

The risk of TdP increases with higher degrees of QT prolongation, typically when the corrected QT interval (QTc) exceeds 500 milliseconds or increases by more than 60 milliseconds from baseline. This risk is not uniform across all Class III agents. For example, dofetilide and ibutilide, which are highly selective IKr blockers, carry a greater risk of TdP compared to amiodarone, which has multiple ion channel effects. Major classes of medication listed below when used with Class III agents can increase QT prolongation risk.

- **Class Ia Antiarrhythmics:** Quinidine, Procainamide, Disopyramide
- **Macrolide Antibiotics:** Erythromycin, Clarithromycin
- **Fluoroquinolones:** Levofloxacin, Moxifloxacin
- **Azole Antifungals:** Ketoconazole, Fluconazole
- **Antipsychotics:** Haloperidol, Ziprasidone
- **Tricyclic Antidepressants:** Amitriptyline

CLINICAL CORRELATION

Sotalol has β-blocking properties but is not classified pharmacologically as a β-blocker.

Adverse Reactions

Of the class III antiarrhythmics, sotalol, dofetilide, and ibutilide can all cause torsades de pointes. Because sotalol is also a β-adrenergic blocker, it can cause bradycardia and bronchoconstriction.

Class III antiarrhythmics, particularly amiodarone, are associated with numerous noncardiac adverse effects that can affect multiple organ systems. Amiodarone's long half-life (25-100 days) and high lipid solubility contribute to its extensive tissue distribution and accumulation, leading to a wide range of toxicities that may persist even after drug discontinuation.

Pulmonary toxicity is one of the most serious noncardiac adverse effects of amiodarone, occurring in approximately 5-10% of patients and manifesting as interstitial pneumonitis, organizing pneumonia, or acute respiratory distress syndrome. Risk factors include higher doses, longer duration of therapy, pre-existing lung disease, and pulmonary angiography. Symptoms include progressive dyspnea, cough, and occasionally fever, with characteristic findings on chest imaging and pulmonary function tests.

Thyroid dysfunction is another common adverse effect of amiodarone, which can cause both hypothyroidism (5-10% of patients) and hyperthyroidism (2-3% of patients). These effects result from amiodarone's high iodine content (37% by weight) and direct toxic effects on thyroid follicular cells.

Other adverse effects bradycardia, peripheral neuropathy, ataxia, and hypotension.

Drug interactions of Class III agents

Class III antiarrhythmics exhibit significant drug interaction profiles. **Amiodarone** inhibits multiple CYP enzymes (CYP1A2, CYP2C9, CYP2D6, CYP3A4) and P-glycoprotein, markedly increasing plasma levels of drugs like **warfarin** (↑50–200%, requiring 30–50% dose reduction and close INR monitoring) and **digoxin** (↑70–100%, dose reduction by ~50%).

Dronedarone, metabolized by CYP3A4, interacts with strong CYP3A4 inhibitors (e.g., ketoconazole, clarithromycin) causing 2–3-fold level increases, and inducers (e.g., rifampin, phenytoin) reducing efficacy.

Sotalol, renally excreted unchanged, avoids CYP interactions but is influenced by drugs altering renal clearance (e.g., metformin, diuretics).

How do the miscellaneous antiarrhythmic agents work? (LO#1–3)

Some antiarrhythmic drugs do not exactly fit into classes I through IV. These miscellaneous agents, adenosine, magnesium, and digoxin are also used as antiarrhythmic drugs.

Adenosine

Adenosine is mainly used to convert acute episodes of AVNRT to sinus rhythm. It works by allowing K^+ to leak out of pacemaker cells, causing a brief hyperpolarization that stops the heartbeat.

Mechanism of Action. This drug slows AV node conduction and interrupts reentry pathways, restoring normal sinus rhythm. The effects of adenosine are short acting (5-15 seconds).

Adverse Effects. Adverse effects include a brief sense of impending doom, bradycardia, bronchospasm, hypotension, chest pain, dyspnea, and flushing.

Magnesium

Magnesium is used as a first-line drug to suppress or terminate torsades de pointes.

Mechanism of Action. It slows the rate of SA node impulse formation in the myocardium and prolongs conduction time. Mg competes with Ca^{2+} decreasing Ca^{2+} influx into the cell, thus decreasing the afterdepolarizations. It also increases K efflux aiding the repolarization to shorten QT durations.

Adverse Effects. The most notable adverse effects are weakness and decreased reflexes when administered at high doses. In fact, upper extremity grip strength and reflexes should be regularly reviewed to monitor for toxicity. It can also produce hypotension, flushing, sweating, and headache due to vasodilation.

Which antiarrhythmic drug can be used to suppress torsades de pointes as a first-line therapy?

Digoxin

Digoxin is uncommonly used for rate control of chronic AF when other options like class II and IV agents have failed. It is generally avoided because of significant side effects and a narrow therapeutic window.

Mechanism of Action. Digoxin primarily acts by increasing vagal tone and slowing conduction through the atrioventricular (AV) node. Enhances parasympathetic (vagal) activity → decreases AV nodal conduction velocity and prolongs AV nodal refractory period.

CLINICAL CORRELATION

Digoxin is a positive inotropic agent used in heart failure. It **inhibits Na^+/K^+-ATPase** → increases intracellular sodium → reduces calcium extrusion via Na^+/Ca^{2+} exchanger → enhances intracellular calcium, improving myocardial contractility (positive inotropy).

Effects on Electrocardiogram. Chronic use of digoxin may cause T wave changes (flattening o and scooped ST segments with ST depression in the lateral leads. These findings are not associated with toxicity or adverse effects.

Adverse Effects. The most frequent cardiac adverse effects of digoxin when supra-therapeutic levels are reached are premature ventricular contractions and a variety of supraventricular, junctional, and ventricular arrhythmias. These can be fatal, which is why this drug is now used less and less.

Toxic levels may also cause confusion, headaches, dizziness, nausea, vomiting, and yellow halos around objects in the visual field.

Putting It All Together

The antiarrhythmic drug classes discussed here are summarized in Table 1.

Table 1 Classes II, IV, and V antiarrhythmic drugs

Class	Target	Examples	Used to treat	Adverse effects
Class II	β-Adrenergic receptors	Metoprolol, propranolol, esmolol, atenolol, timolol, carvedilol	Atrial or ventricular arrhythmias	Erectile dysfunction, bradycardia, bronchoconstriction; masks hypoglycemia
Class IV	L-type Ca^{2+} channels	Verapamil, diltiazem	Rate control of atrial arrhythmias	Constipation, peripheral edema, bradycardia, negative inotropy
Class III	K^+ channels	Amiodarone, Ibutalide, Dronadarone	Rate and rhythm control	QT prolongation, bradycardia, drug interactions and hepatotoxicity (with Amiodarone), Pulmonary fibrosis darone)
Miscellaneous	Variable	Adenosine, magnesium, digoxin	Adenosine: AVNRT Magnesium: torsades de pointes, digoxin toxicity Digoxin: third line to slow atrial fibrillation	Adenosine: sense of doom, bronchospasm, hypotension, chest pain, flushing; augmented by caffeine and theophylline Magnesium: weakness, decreased reflexes, hypotension, flushing, sweating, headache Digoxin: mental disturbances, nausea, vomiting, yellow halos, dizziness, headache

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to JG, what is the mechanism of action of verapamil in the treatment of his atrial fibrillation? What screening questions would you ask him before prescribing this medication?

You recall that calcium-channel blockers (CCBs) are useful in the treatment of atrial fibrillation as they prolong depolarization and repolarization primarily at the atrioventricular (AV) node. The result is slowing of the heart rate. Because calcium is also required for bowel peristalsis, CCBs can also cause constipation. You suggest stopping verapamil and starting a β-blocker, which will slow sinoatrial and AV node conduction velocity after making sure that JG has no history of bronchospastic lung disease (asthma or COPD.) JG agrees to start metoprolol, a commonly used β-blocker, and does well in follow-up.

Summary

- Class II antiarrhy hetic influence on the heart, slowing SA and AV node conduction and treating supraventricular arrhythmias.
- Class IV antiarrhythmics, such as verapamil and diltiazem, block L-type calcium channels, slowing depolarization and repolarization in nodal cells, used for supraventricular tachycardias.
- Class III antiarrhythmics block K+ channels, prolonging repolarization and increasing action potential duration, used for atrial and ventricular arrhythmias.
- Miscellaneous antiarrhythmics include adenosine, magnesium, and digoxin, which have unique mechanisms for treating specific arrhythmias like AVNRT and torsades de pointes.
- Adverse effects of β-blockers (Class II) and CCBs (Class IV) include bradycardia, bronchospasm, and masking hypoglycemia signs, while calcium-channel blockers may cause constipation and edema.
- Class III agents can cause QT prolongation and torsades de pointes, with amiodarone associated with pulmonary toxicity and thyroid dysfunction.

Review Questions

1. A patient is started on verapamil for an atrial fibrillation. Which of the following correctly describes this drug's mechanism of action?

A. ● Blocks calcium channels
B. ○ Blocks potassium channels
C. ○ Blocks sodium channels with strong use dependence
D. ○ Blocks sodium channels with weak use dependence
E. ○ Inhibits sympathetic input to cardiac myocytes

Hide Explanation

The correct answer is blocks calcium channels (A). Verapamil and diltiazem are class IV antiarrhythmics that work predominantly on nodal cells. Blocks potassium channels (B) is the action of class III antiarrhythmics. Blocks sodium channels with strong use dependence (C) describes class IC antiarrhythmic agents such as propafenone and flecainide. Blocks sodium channels with weak use dependence (D) describes Class IB antiarrhythmics like lidocaine and mexiletine. Inhibits sympathetic input to cardiac myocytes (E) describes β-blockers.

2. Which of the following drugs produces yellow halos as a drug toxicity?

A. ○ Adenosine
B. ● Digoxin
C. ○ Diltiazem
D. ○ Magnesium
E. ○ Verapamil

Hide Explanation

The correct answer is digoxin (B). Patients taking digoxin may experience mental disturbances, nausea, vomiting, yellow halos, dizziness, and headaches. Adenosine (A) may produce bronchospasm, hypotension, chest pain, flushing, and a sense of impending doom. Diltiazem (C) is a calcium-channel blocker that may precipitate constipation, peripheral edema, and bradycardia. Patients treated with magnesium (D) may have weakness, decreased reflexes, hypotension, flushing, sweating, and headaches. Verapamil (E) is another calcium-channel blocker.

3. A 68-year-old man with a history of atrial fibrillation has been on antiarrhythmic therapy for 8 months. He presents with progressive shortness of breath, nonproductive cough, and low-grade fever. Physical examination reveals fine inspiratory crackles. Chest X-ray shows bilateral interstitial infiltrates, and pulmonary function tests demonstrate a restrictive pattern with reduced diffusion capacity. He denies chest pain or hemoptysis.

A. ● Amiodarone
B. ○ Sotalol
C. ○ Flecainide
D. ○ Dofetilide

Hide Explanation

The correct answer is Amiodarone (A). Amiodarone is strongly associated with **pulmonary toxicity** (interstitial pneumonitis, organizing pneumonia), especially after prolonged use. Risk factors include high cumulative dose and pre-existing lung disease. Sotalol (B) is a Class III agent that prolongs QT interval but does not cause pulmonary toxicity; its major risks are torsades de pointes and bradycardia. Flecainide (C) is a Class Ic agent that can cause proarrhythmia and conduction abnormalities but does not affect lung tissue. Dofetilide (D) is another Class III drug with risk of QT prolongation and torsades, but no significant pulmonary adverse effects.

4. A patient with a history of asthma is brought to the emergency department with complaints of palpitations and dizziness. The patient is evaluated, and an ECG shows atrial fibrillation. She is then started on an antiarrhythmic drug, after which she develops shortness of breath and wheezes over bilateral lung fields. Which of the following drugs is most likely to have caused this adverse effect?

A. ○ Digoxin
B. ○ Diltiazem
C. ○ Magnesium
D. ● Propranolol
E. ○ Verapamil

Hide Explanation

The correct answer is propranolol (D). Propranolol is a class II antiarrhythmic agent (ie, a β-blocker), exerting the antiarrhythmic effect by inhibiting sympathetic influence on the heart. These agents are used to treat supraventricular arrhythmias such as atrial fibrillation as well as ventricular arrhythmias. However, nonselective β-blockers, such as propranolol, cause $β_2$ antagonism leading to bronchial smooth muscle contraction, precipitating asthma or chronic obstructive pulmonary disease. Digoxin (A) is a "class V" antiarrhythmic agent with toxicities like mental disturbances, yellow halos, and dizziness. Diltiazem (B) is a class IV antiarrhythmic agent that works on L-type Ca^{2+} channels, with major side effects being constipation, peripheral edema, and bradycardia. Magnesium (C) is another class V antiarrhythmic agent with side effects such as weakness, decreased reflexes, hypotension, and flushing. Verapamil (E) is another calcium-channel blocker.

5. Which drug is most likely to increase the refractory period by inhibiting Na^+ and K^+ channels during a cardiac action potential?

A. ● Amiodarone
B. ○ Digoxin
C. ○ Atropine
D. ○ Verapamil

Hide Explanation

The correct answer is Amiodarone (A). Amiodarone is a Class III antiarrhythmic that blocks both potassium (K^+) channels and sodium (Na^+) channels, prolonging repolarization and increasing the effective refractory period during the cardiac action potential. This helps prevent re-entry arrhythmias and stabilizes rhythm. Digoxin (B) acts mainly by increasing vagal tone and slowing AV nodal conduction; it does not significantly affect Na^+ or K^+ channels to prolong refractory period. Atropine (C) is a muscarinic antagonist that increases heart rate by reducing parasympathetic tone; it has no direct effect on cardiac ion channels or refractory period. Verapamil (D) is a calcium channel blocker that slows AV nodal conduction and decreases heart rate; it does not inhibit Na^+ or K^+ channels or prolong refractory period.

6. Which agents is most likely to increase PR interval ?

A. ○ Digoxin
B. ○ Adenosine
C. ○ Atropine
D. ● Metoprolol

Hide Explanation

The correct answer is D, Metoprolol. Metoprolol is a betaβ-1 selective blocker that decreases AV nodal c lation. This slows impulse transmission ion of the PR interval on ECG. Digoxin (A) increases vagal tone and slows AV nodal conduction, which can also prolong PR interval, but its primary effect is rate control in atrial fibrillation rather than consistent PR prolongation like beta-blockers. Adenosine (B) causes transient AV nodal block and can briefly increase PR interval, but its effect is very short-lived and not sustained. Atropine (C) is a muscarinic antagonist that increases heart rate and shortens PR interval by reducing vagal tone, opposite of the effect in question

How would you rate this Brick?

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Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Explain the mechanism of action of antiarrhythmic agents (Class II, III, and IV) and digoxin.~~
02 ☑ ~~Explain the changes in action potential and the associated ECG changes with digoxin and Class II, III, IV antiarrhythmic agents.~~
03 ☑ ~~Recognize the adverse effects of antiarrhythmic agents (Class II, III, and IV) and digoxin.~~

Objective Confidence: 3/3